CLUSTERING

Outline

- Discuss concept of clustering and common applications
- Examine steps in clustering
- Discuss four clustering algorithms, hierarchical clustering, k-means, gaussian mixture models, and density based spatial clustering.

Cluster Analysis

- Unsupervised machine learning technique to group a set of objects into clusters based on their similarity
- Objects in the same cluster are more similar to each other than to those in other clusters
- Similarity is assessed based on a set of relevant features

We are all accustomed to organizing our belongings.

Clustering is about organizing data.

Characteristics

- Unsupervised learning technique
- Widely used in exploratory data analysis to uncover hidden patterns or structures within data.
- Does not generate a unique solution. Cluster solution is chosen is by combining algorithm results with domain knowledge.

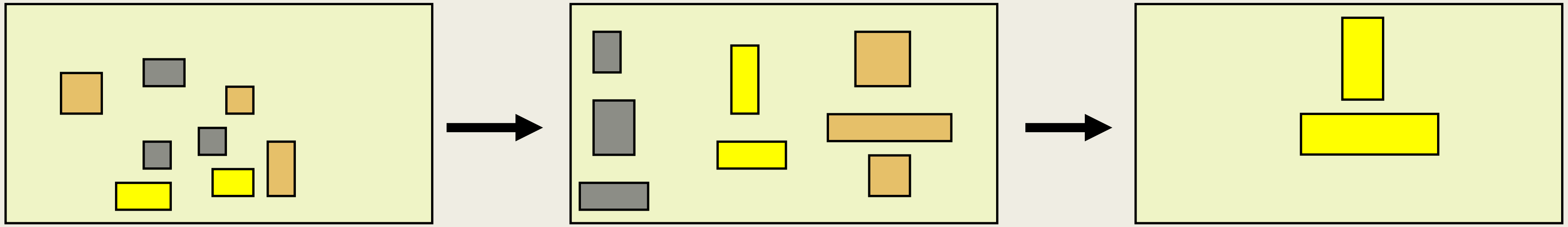
Applications

- Market Segmentation
  - *Group customers based on there attitudes and buying behavior*
- Retail
  - *Group products based on purchase pattern*
  - *Fraud detection based on transaction history*
- Financial Services
  - *Group customers based on financial products or risk profiles*
  - *Anomaly detection to spot fraud*
  - *Group stocks based on performance*
  - *New trends in time series*
- Healthcare
  - *Group patients based on symptoms*
- Image segmentation
  - *Organizing images (e.g., for object identification in computer vision)*
- Text
  - *Group customer reviews*
- Input to predictive modeling
  - *Cluster then predict approach*

Application: Market Segmentation

- Customers are happier when they get what they like but the cost of personalization is very high. Segmentation strikes a balance by identifying groups of customers with similar needs.

Place customers into homogeneous clusters and select a segment to target



Cluster Analysis of Needs

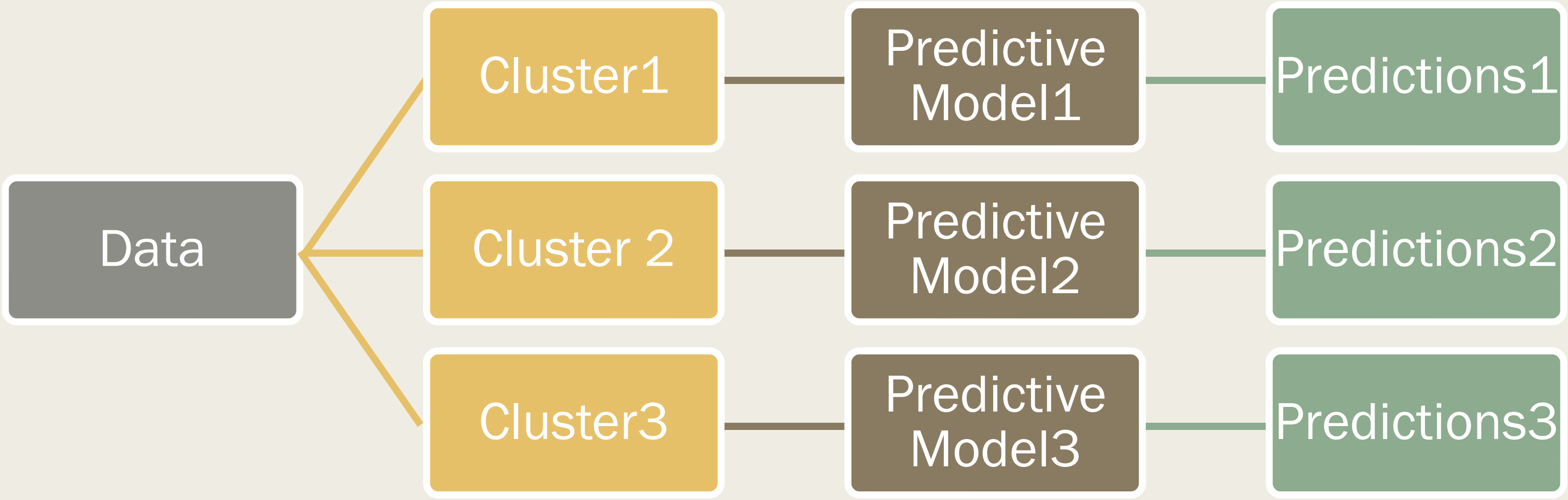
Profile Clusters

Application: Market Segmentation

- Heuristic Approach
  - *Use observable variables to segment the market*
  - *Variables selected to segment are expected to be the ones on which customer needs vary*
  - *This is the traditional approach and does not involve cluster analysis*
- Needs-based Clusters
  - *Use data on customer needs to generate clusters of individuals that differ in their needs*
  - *Profile clusters on observable variables (e.g., gender, age, location)*

Application: Cluster then Predict

- Clustering data before predictive modeling can improve prediction quality
- Cluster Data – Predictive Model for each Cluster – Combine Predictions



PROCESS OF CLUSTER ANALYSIS

Steps in Cluster Analysis

1. Select variables
2. Prepare data
3. Select metric for similarity
4. Select clustering analysis technique
5. Determine number of clusters
6. Interpret Clusters
7. *If labels are available, evaluate clusters*
8. *If another sample is available, validate clustering scheme*

Select variables

- Clusters are only as good as the variables chosen to define it.
- Software cannot offer guidance on the practical use of variables.
- Consider relevance of variables chosen to the problem
  - *In market segmentation, variables are chosen based on customer needs. For a fitness club, this could include, visit frequency, and workout goals.*
  - *On the other hand, is number of visits/week to the gym relevant in segmenting the market for Folgers coffee? What about variety seeking behavior? What about innovativeness?*
- Use of multiple variables tapping the same dimension may overweight the dimensions' impact in defining clusters.
- Nature of variables selected influence choice of clustering algorithm

Prepare Data

- Transform format of variables
  - *Clustering algorithms prefer all variables to be of the same class (e.g., numeric, factor)*
- Impute missing data
  - *Clustering algorithms use data on all variables. A missing observation on one variable will cause the entire row of data to be ignored for analysis. Therefore, it is important to impute missing values.*
- Scale
  - *For distance-based clustering methods, scale of the variable (e.g., seconds vs minutes) affects the weight assigned to the variable. Therefore, it is best to standardize variables.*
- Redundant variables
  - *Multiple variables measuring the same dimension should be replaced by the underlying factor or a representative variable.*

Metric for Similarity

- Distance between observations
- Density of data
- Underlying distribution for data

Select Cluster Analysis Technique

- Hierarchical Clustering
  - *Hierarchical Approach that builds a tree-like structure of nested clusters.*
- k-means Clustering
  - *Partitioning Approach that divides data into k clusters to minimize within-group dissimilarity*
- Gaussian Mixture Models
  - *Model-Based Approach that assumes data is generated from a mixture of underlying distributions.*
- DBSCAN
  - *Density-Based Spatial Clustering of Applications with Noise (DBSCAN) is a density-based approach that groups regions of high density separated by low-density regions*

Number of Clusters

- Domain knowledge
  - *Whenever possible, the decision of number of clusters must be based on an understanding of the problem.*
- Visual inspection
  - *With techniques such as hierarchical clustering, the number of clusters can be inferred from dendrogram*
  - *For k-means, an elbow plot and silhouette plot indicate optimal number of clusters*
- Statistical criterion
  - *Metrics such as average silhouette width (for k-means) and BIC (for gaussian mixture models*

Interpret Clusters

- Examine the profile of each clusters based on features used for clustering (e.g., needs-based variables used for market segmentation) as well as variables not used for clustering (e.g., demographics for a market segmentation problem).

HIERARCHICAL CLUSTERING

Hierarchical Clustering

- Popular method that groups observations according to their similarity
- Distance-based algorithm that operates on a dissimilarity matrix
- Algorithm begins with each observation in its own cluster.
- Successively joins neighboring observations or clusters one at a time according to their distances from one another, and continues until all observations are linked
- Process of repeatedly joining observations is called the agglomerative method. Divisive method works in the reverse order.

Hierarchical Clustering

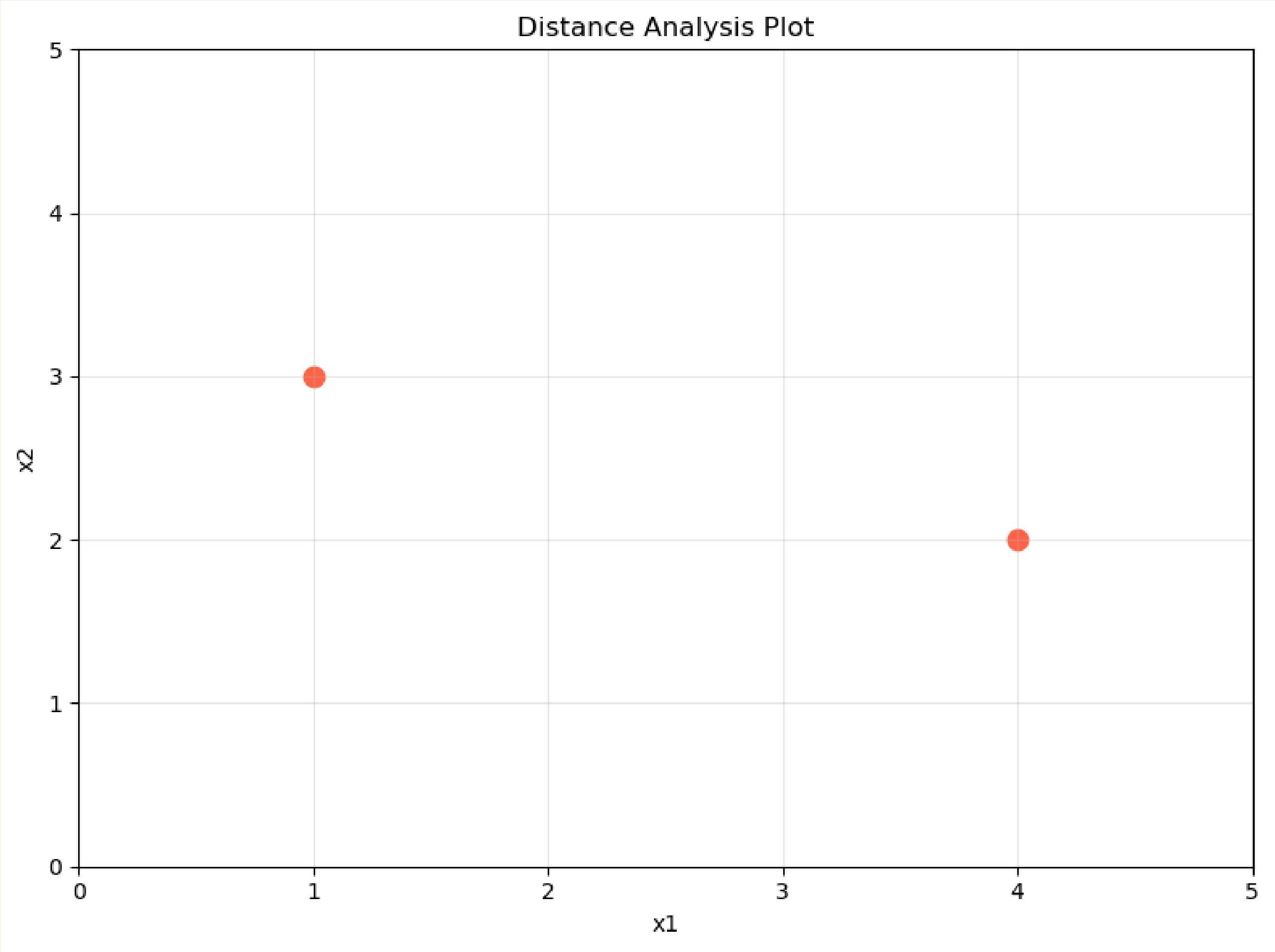
- Hierarchical clustering builds a tree-like structure (dendrogram) of nested clusters using either an agglomerative (bottom-up) or divisive (top-down) approach.

- Strengths:
  - *Reveals hierarchical relationships between clusters.*
  - *Does not require the number of clusters a priori.*
- Weaknesses:
  - *Computationally expensive for large datasets.*
  - *Sensitive to noise and outliers.*

Distance Measure

- Common distance measures:
  - *euclidean distance*
  - *standardized euclidean distance*
  - *cityblock or Manhattan*
  - *minkowski distance*
  - *chebyshev distnace*

Distance Measure



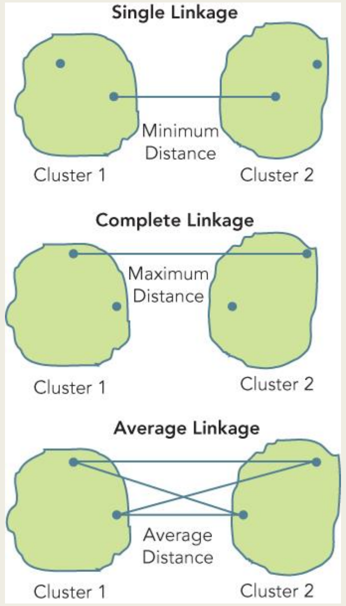
Method	Distance
euclidean	3.162278
standardized euclidean	2.000000
cityblock or manhattan	4.000000
minkowski	3.162278
chebyshev	3.000000

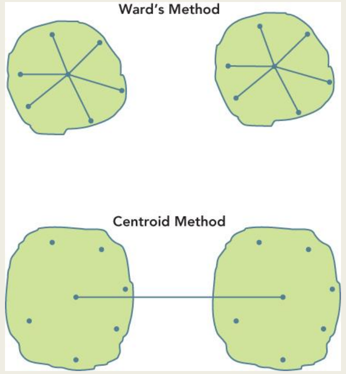
Clustering or Linkage Method

- Distance between pairs of points is useful in identifying similarity between a pair of points.
- As observations are combined into clusters, distances must now be evaluated among clusters or between a point and a cluster.
- There are different approaches to grouping observations and these are called clustering methods.

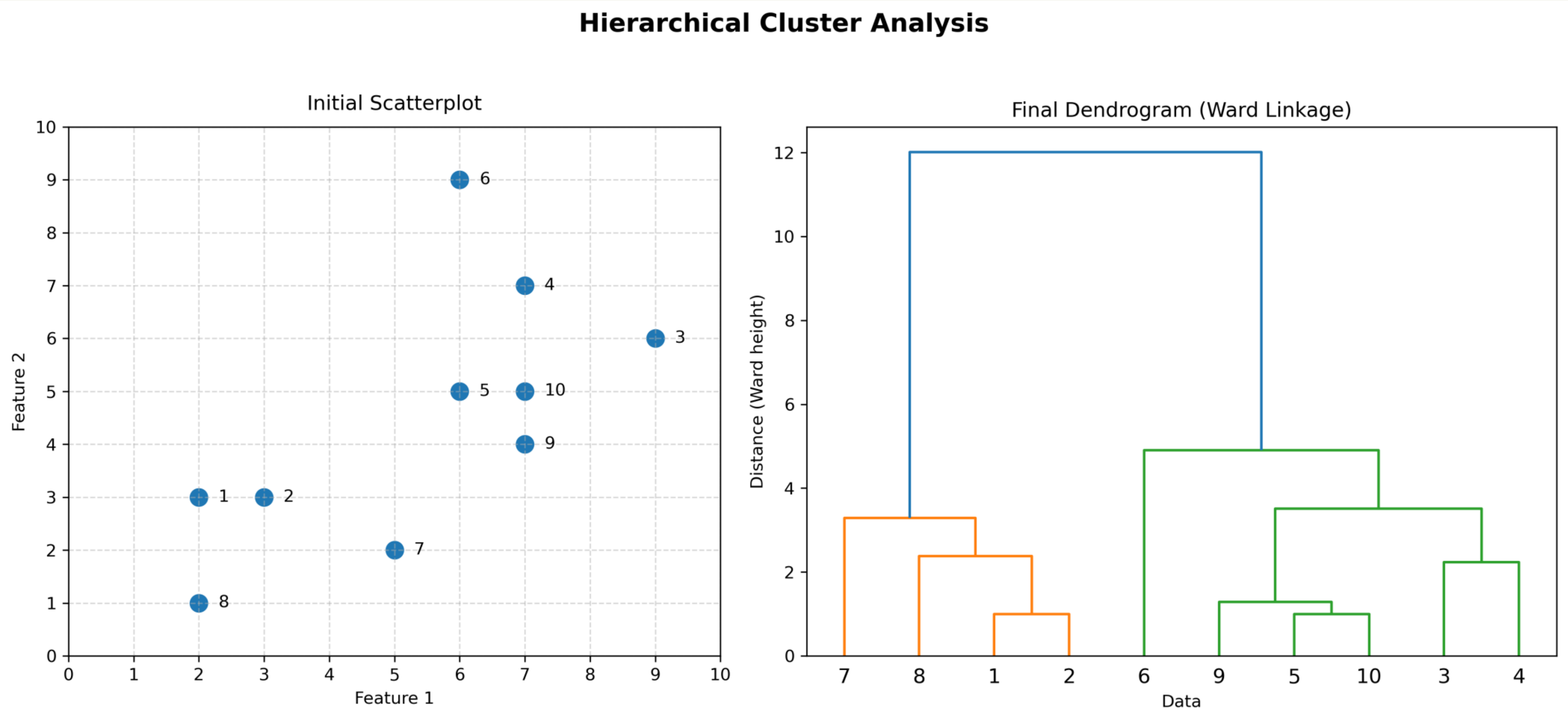
Clustering or Linkage Method

- Linkage methods generate clusters based on distance between observations
  - *single*
  - *complete*
  - *average*
- Variance methods generate clusters to minimize the within-cluster variance. Ward's minimum variance method is aimed at finding compact spherical clusters
  - *ward*
- Centroid methods compute distance between cluster centroids
  - *median*
  - *centroid*



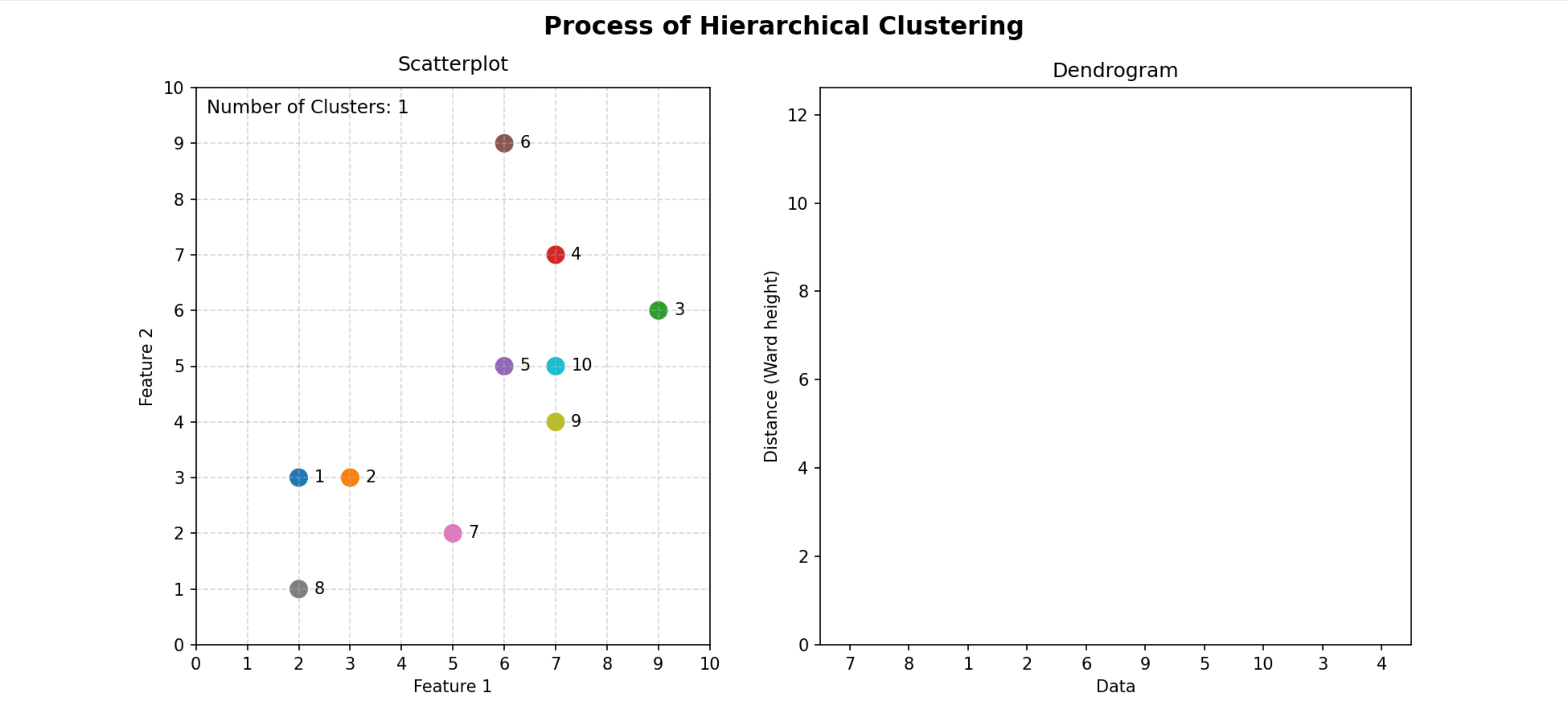


Process



Hierarchical Cluster Analysis
Initial Scatterplot
Feature 1
Feature 2
0
1
2
3
4
5
6
7
8
9
10
Final Dendrogram (Ward Linkage)
Distance (Ward height)
Data
0
2
4
6
8
10
12
7
8
1
2
6
9
5
10
3
4

Process Animated



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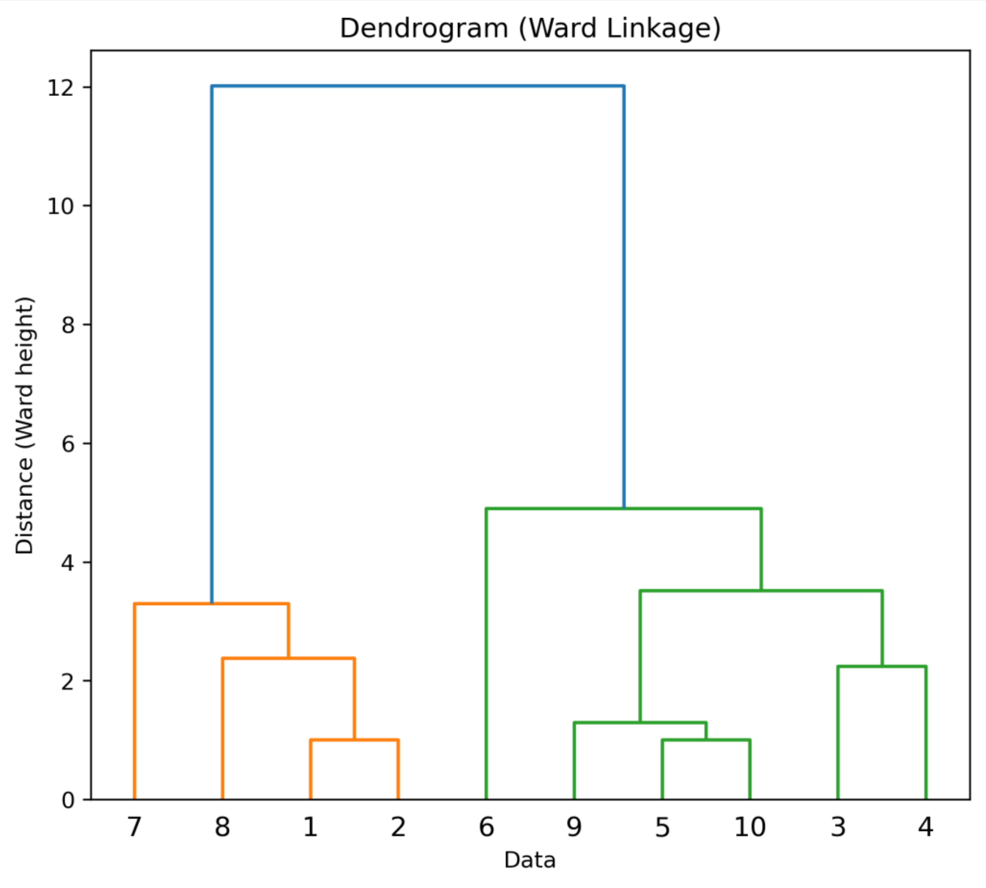
Determine Number of Clusters

- Number of clusters may be determined by examining the height between branch splits.
- Cut the dendrogram to yield clusters that are furthest apart.



Meaningfulness of Clusters

- Hierarchical clustering will converge on a single cluster solution.
- Ultimate decision on the number of clusters to pick should be made based on distances and meaningfulness of clusters
- The characteristics of the clusters should be examined based on the features used for clustering and other variables not used for clustering.
- In conducting market segmentation, it is common to segment based on needs-based variables (e.g., attitude, preference, and product usage), but profile segments based on demographic variables (e.g., age, gender, income).

Limitations

- Hierarchical clustering is simple and intuitive. However, this technique does not scale well to large datasets. This is because, the technique requires computing distances between every pair of observations. For large datasets, this can be a very expensive process, (for n observations there are n(n-1)/2 distances).
- The computation of the distance matrix can consume significant computing resources and the distance matrix itself can overwhelm system memory.
- Let us turn to another technique that is less resource intensive.

K-MEANS CLUSTERING

k-means Clustering

- Attempts to find groups that are most compact, in terms of the mean sum-of-squares deviation of each observation from the multivariate center (centroid) of its assigned group
- Non-hierarchical process that reaches solution through an iteration.
- K-means clustering begins by arbitrarily placing centroids in the data and then iterating from that point to the final solution. This disorganized approach to clustering produces similar quality of clusters to hierarchical clustering but much faster.

k-Means Clustering

- Strengths
  - *Fast, simple, and scalable.*
  - *Performs well with spherical and uniformly distributed clusters.*
- Weaknesses
  - *Requires the number of clusters (k) to be specified.*
  - *Sensitive to initialization and outliers.*
  - *Performs poorly when the clusters are not spherical (e.g., elliptical).*
    - Does not do well with imbalanced clusters.

k-means Process

- Requires an a priori definition of number of clusters
- Begins with an arbitrary assignment of observations to cluster centroids. Seed is important for reproducibility.
- Iterative process involves
  - *Updating cluster centers*
  - *Reassigning observations*
- Updating is done to minimize sum of squares from observations to cluster centers
- Since it explicitly computes a mean deviation, k-means clustering relies on Euclidean distance
- It is only suitable for numerical data
- Distance is computed using Euclidean's method
- Method of updating is defined by algorithms including
  - *"lloyd", "elkan"*

k-means

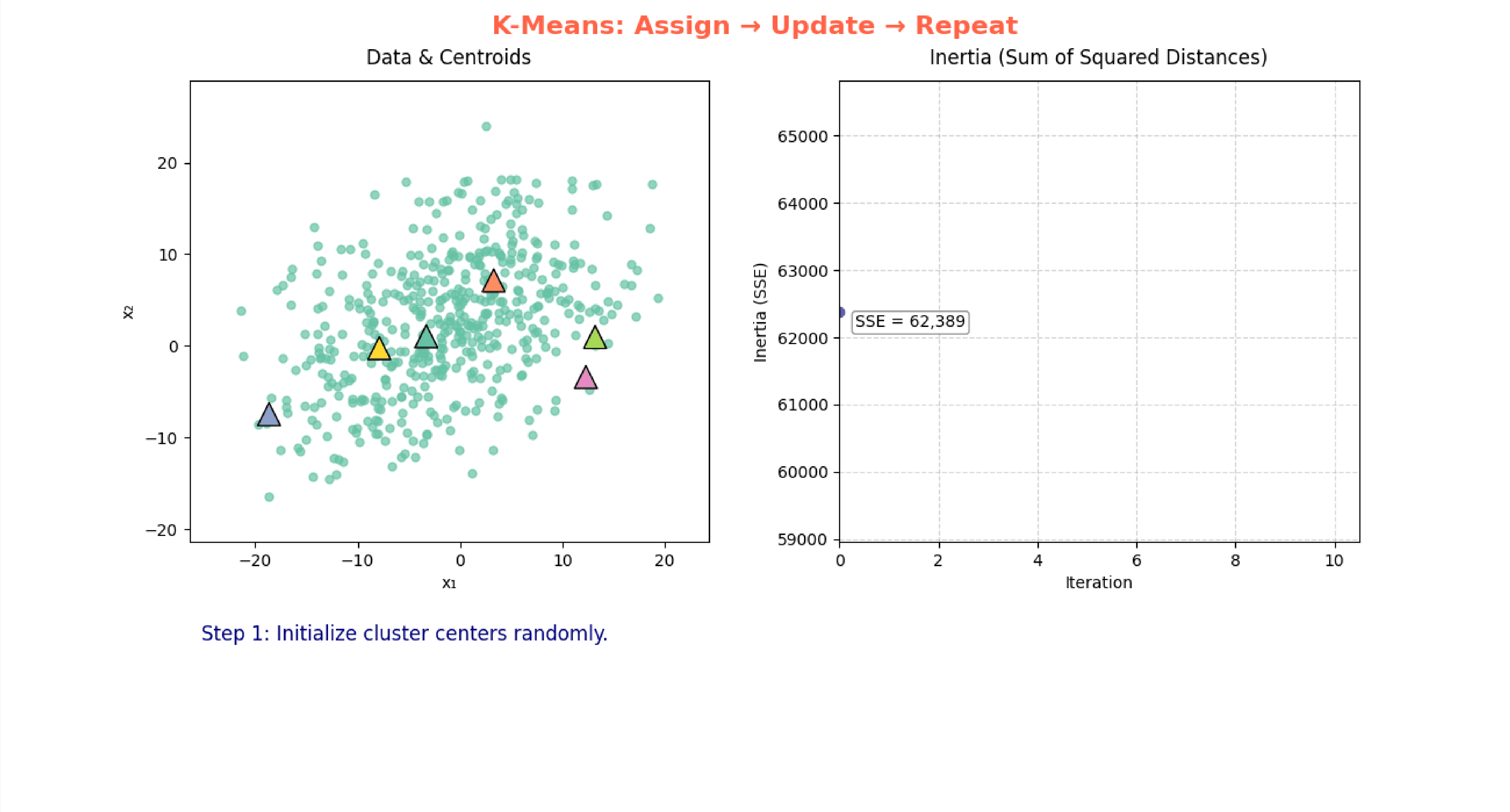
- Initial assignment of centers is influenced by seed or random state
- Sensitive to scale of variables as similarity is based on Euclidean distance

Visualizing k-means

k-means clustering algorithm involves the following steps

1. Identify number of centers, k, to be used
2. Randomly place centers in data
3. Assign observations to nearest center
4. Update cluster centers
5. Reassign cases
6. Repeat steps 4 and 5 until convergence

Next few slides visually illustrate the process until convergence is reached in iteration 5.



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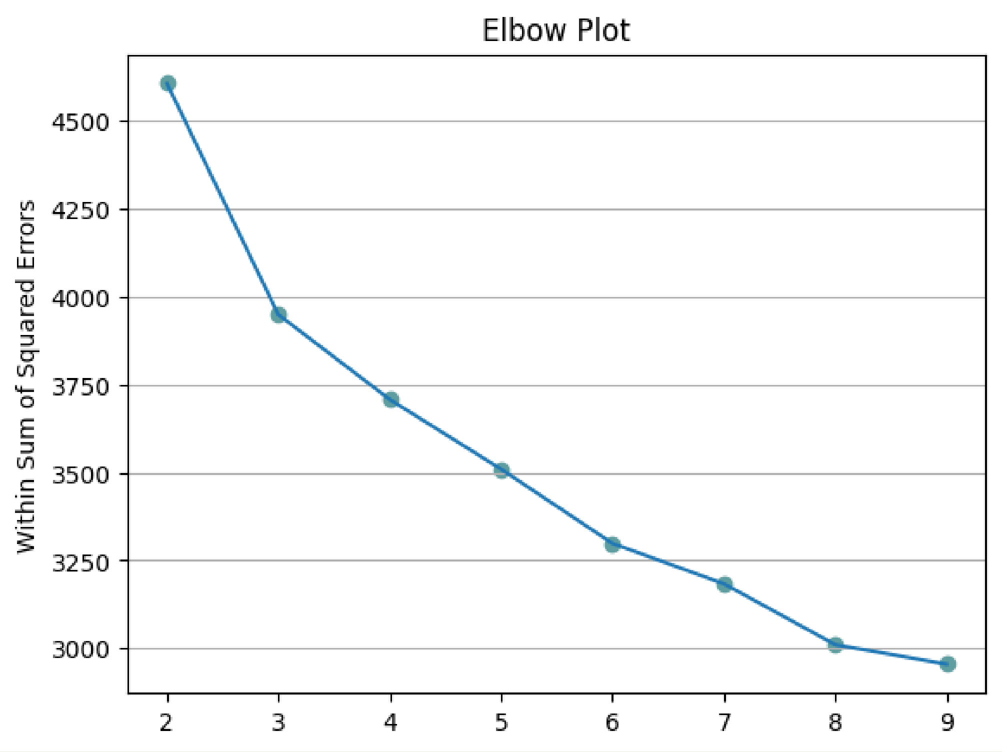
k: Number of Clusters

- For k-means, the decision of k must be made before running the algorithm
- Initial choice of k must be well thought-out and rely on domain knowledge as it drives the cluster solution
- There are also a few data-driven approaches to determining clusters. Two of them are
  - *Total within cluster sum of squares*
  - *Silhouette Method*

k: Number of Clusters
Within sum of squares

- Total within sum of squares Plot
  - *Compute total within sum of squares for a number of values of k. Plot a line graph of k (on x-axis) against total within sum of squares (on y-axis). Ideal number of clusters is inferred from a sudden change in the line graph or what is commonly known as the "elbow".*
- Ratio Plot
  - *Another alternative that generates a similar conclusion is to compute the ratio of between cluster sum of squares and total sum of squares for a number of values of k. Ideal number of clusters is inferred from the elbow in the graph.*

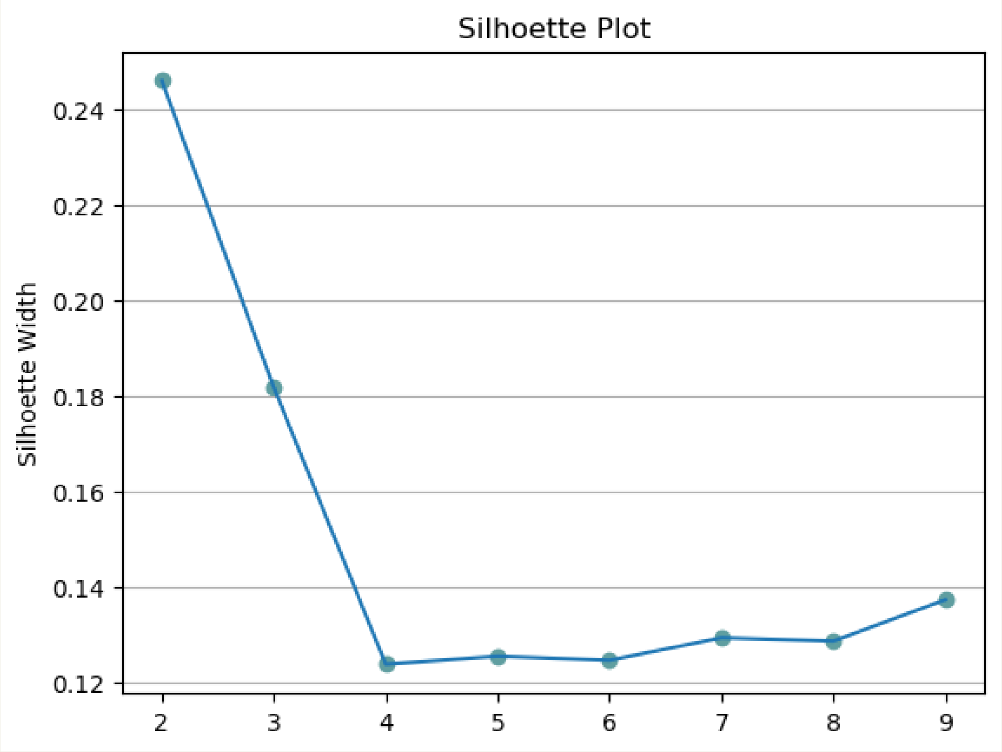
k: Number of Clusters
Within sum of squares



k: Number of Clusters Silhouette Method

- Silhouette analysis reflects how well points fit in their respective clusters
- It involves computing the silhouette width(s(i)) for each point
- s(i) = (b(i) – a(i))/max{a(i),b(i)}
  - *where*
    - a(i) is average distance of a point to all observations within its cluster
    - b(i) is the average distance of a point to all observations in the nearest cluster
- -1 <= s(i) <= 1
  - *s(i) = 1 implies i is in the right cluster*
  - *s(i) = 0 implies i is between two clusters*
  - *s(i) = -1 implies i should be in the nearest cluster*

k: Number of Clusters Silhouette Method

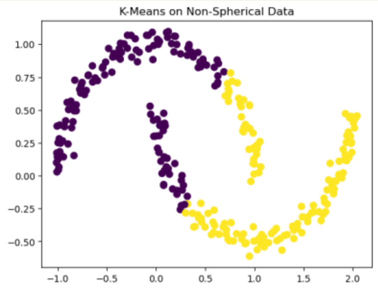


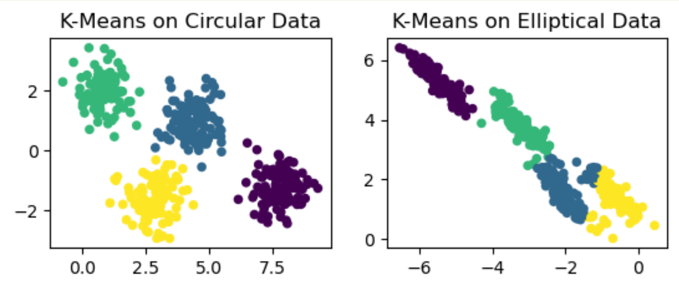
Strengths

- Fast, simple, and scalable
- Relatively low computational power over hierarchical clustering
- Performs well with spherical and uniformly distributed clusters.

Limitations

- Requires the number of clusters (k) to be specified. Lacks the benefit of a dendrogram for inferring clusters
- Sensitive to starting values, which are based on the seed
- Sensitive to outliers
- Does not do well with imbalanced clusters
- Only allows the use of one distance metric – Euclidean
- Performs poorly when the clusters are not spherical (e.g., elliptical)

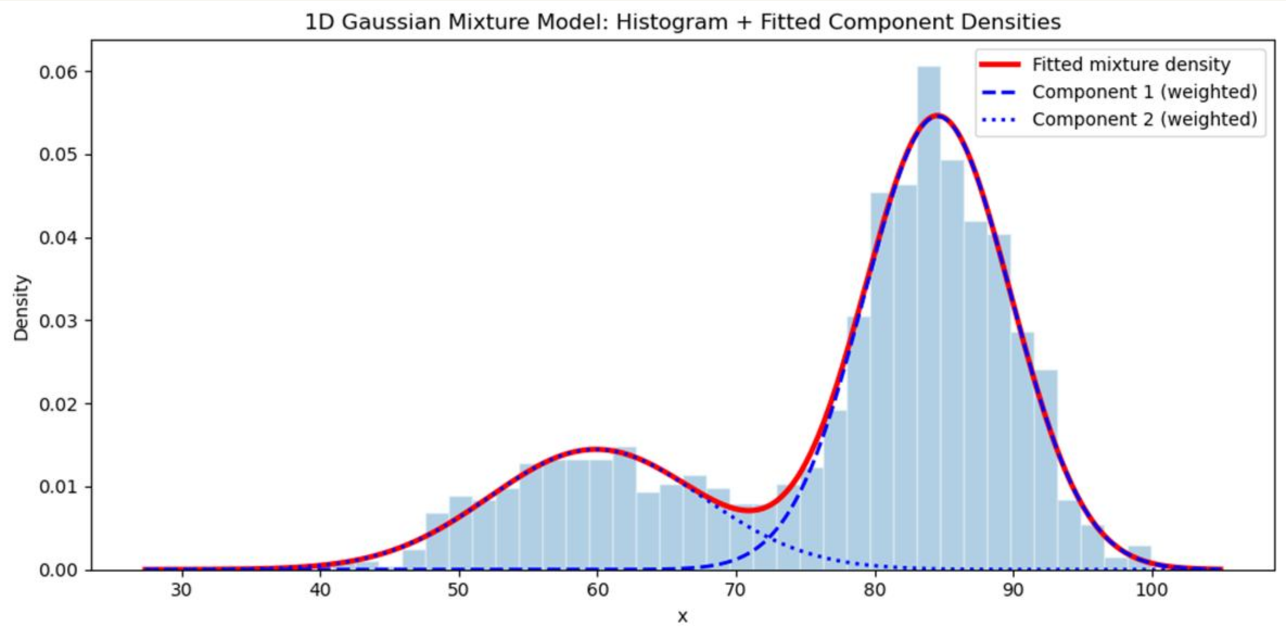




GAUSSIAN MIXTURE MODELS

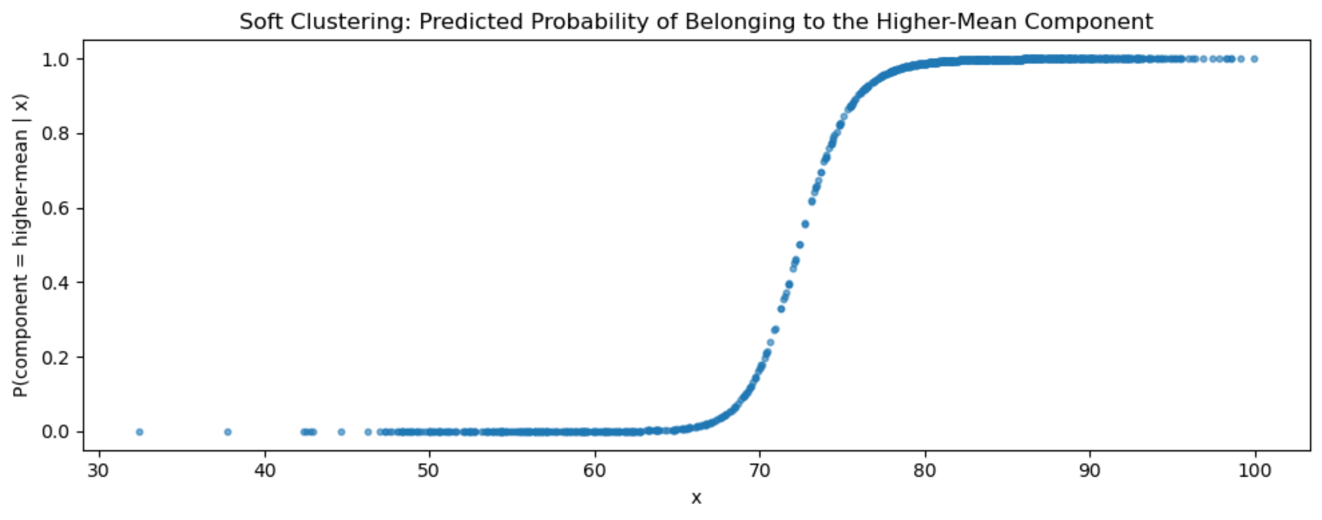
Gaussian Mixture Models

- Key idea of model-based clustering is that observations come from groups with different statistical distributions (such as different means and variances). The algorithms try to find the best set of such underlying distributions to explain the observed data.
- These methods are also known as mixture models because it is assumed that the data reflect a mixture of observations drawn from different populations.



Soft Clustering

- Since these models contain a probabilistic model under the hood, it is possible to find probabilistic cluster assignments.

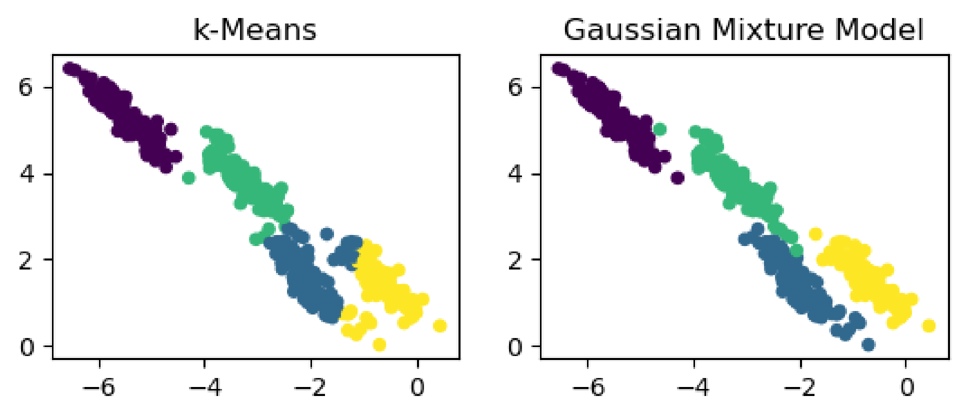


Iterative Process

- These models use an expectation maximization approach to iterate to the optimal solution. Since this approach is iterative, and likelihood-based, the algorithm is not guaranteed to find the global maximum. As a result, starting values can affect results.
- Process
  - *Choose Starting Guesses for the location and shape*
  - *Repeat until convergence*
    - E-step: For each point, find weights encoding the probability of membership in each cluster
    - M-step: For each cluster, update its location, normalization, and shape based on all data points, making use of the weights

Strengths

- Performs soft clustering (points can belong to multiple clusters).
- Models overlapping clusters effectively.
- Works well on complicated geometrical shapes

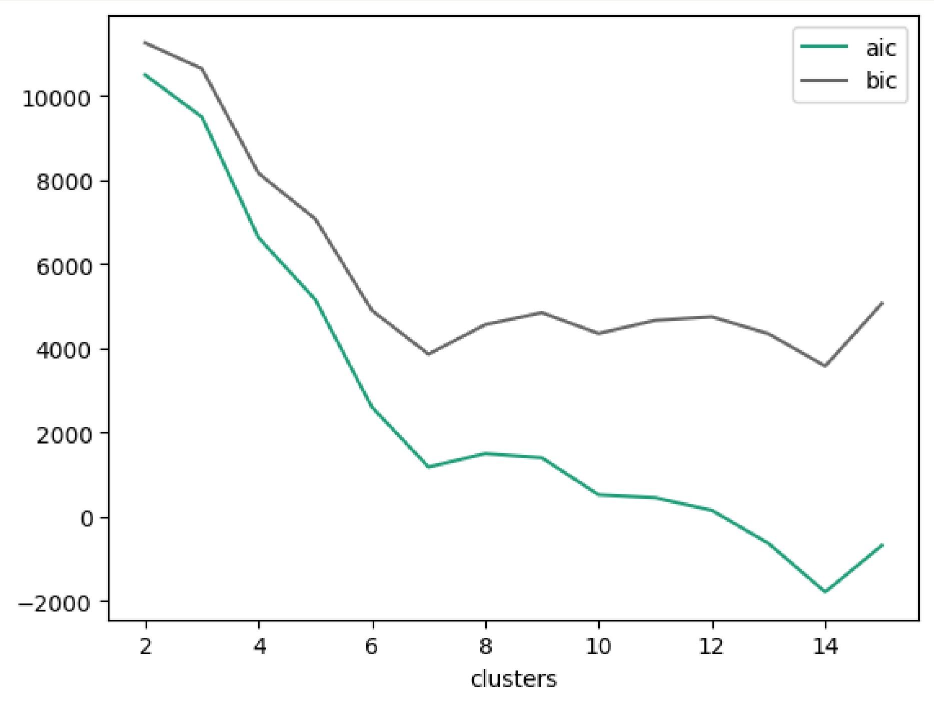


Weaknesses

- Sensitive to initialization.
- Computationally expensive for high-dimensional data.

Number of Clusters

- Choice of clusters is usually based on log-likelihood adjusted for overfitting using criterion like AIC or BIC.



DBSCAN

DBSCAN

Density-based spatial clustering of applications with noise

- Density-based clustering algorithm that groups together points that are closely packed, marking points that lie alone in low-density regions as outliers or noise. Unlike other clustering algorithms, DBSCAN can discover clusters of arbitrary shapes and is particularly effective for identifying outliers.

DBSCAN

- Creates clusters centered around spatial centroids.
- Needs to be provided density value.
- Area immediately around the centroid is referred to as a neighborhood, and DBSCAN attempts to define neighborhoods of clusters with a specified density.
- Can discover clusters of any size, shape or density.
- Can distinguish observations that should be part of a cluster and noise. Thus, it is particularly useful in the presence of outliers and when number of clusters is not known.

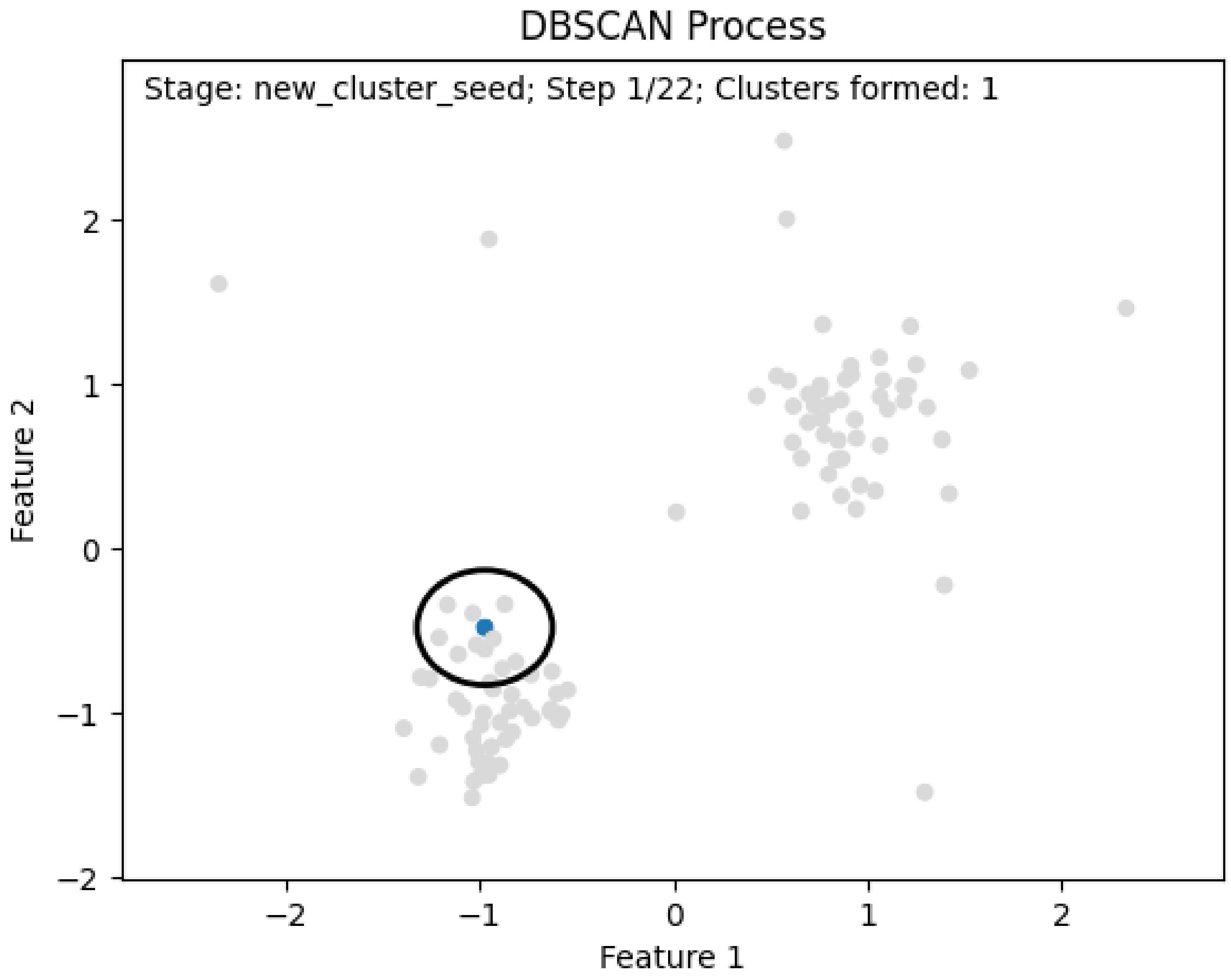
Algorithm

- For each instance, the algorithm counts how many instances are located within a small distance ε (epsilon) from it. This region is called the instance's ε-neighborhood.
- If an instance has at least min_samples instances in its ε-neighborhood (including itself), then it is considered a core instance. In other words, core instances are those that are located in dense regions.
- All instances in the neighborhood of a core instance belong to the same cluster. This neighborhood may include other core instances; therefore, a long sequence of neighboring core instances forms a single cluster.
- Any instance that is not a core instance and does not have one in its neighborhood is considered an anomaly.

Process

- Set epsilon and min_samples
  - *epsilon: maximum distance between two points for them to be considered part of the same neighborhood.*
  - *min_samples (or MinPts): number of points in the neighborhood*
- Every data point is labeled as
  - *Core point: If it has at least min_samples points within the epsilon neighborhood*
  - *Border point: Non-core points within a core neighborhood*
  - *Noise point: Isolated from all core-point neighborhoods*
- Clusters: Core points + Border points
- Noise: Non-clustered points

Visualization of Process



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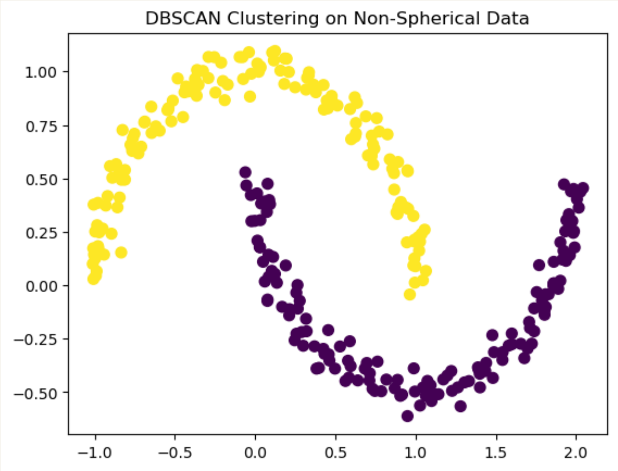
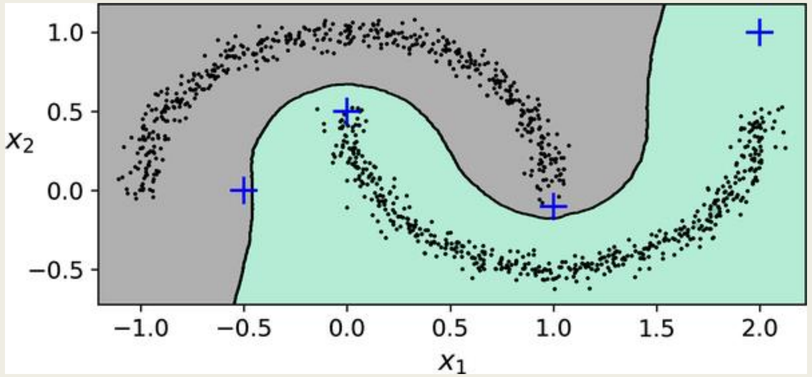
Other features

- Not iterative
- Grows clusters in one pass without updating them once they are labeled
- Unassigned points are treated as noise
- Hierarchical DBSCAN, an extension of DBSCAN is able to generate density-based clusters without needing eps to be set.

Strengths

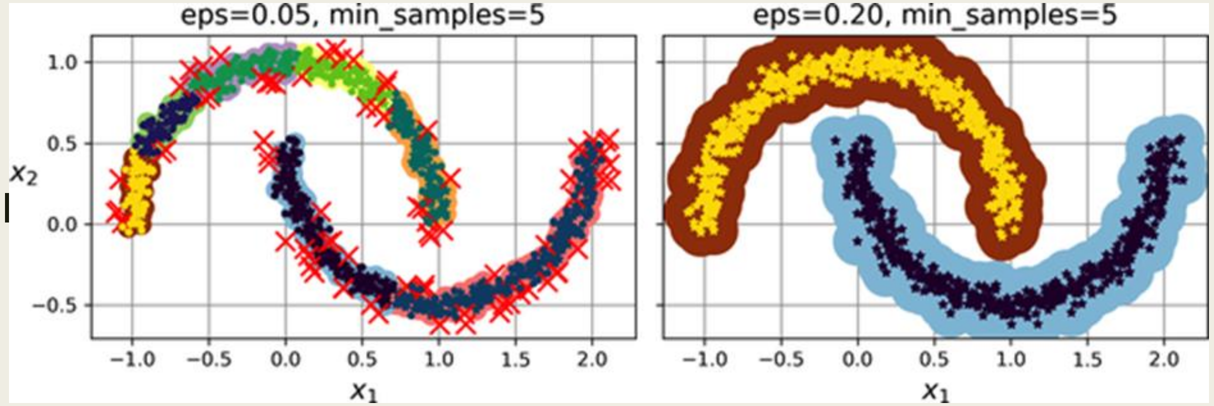
- Identifies clusters of any shape
- Automatic Outlier Detection
- Robust to noise and outliers





Weaknesses

- Sensitive to parameter selection (e.g., eps, min_samples)
- Struggles with clusters of varying density. Specifically, it struggles if there is no low-density area around clusters
- Parameter values optimized based on average silhouette width may not yield meaningful clusters.



Conclusion

- In this module, we
  - *discussed the concept of clustering*
  - *examined the steps in clustering*
  - *discussed four clustering algorithms, hierarchical clustering, k-means, gaussian mixture models, and density based spatial clustering*